

5. Hypergeometric Distribution

To obtain a formula analogous to that of the binomial distribution that applies to sampling without replacement, in which case the trials are not independent.

Let us consider a set of N elements of which M are looked upon as successes and the other $N-M$ as failures. As in connection with binomial distribution, we are interested in the probability of getting x successes in the trials, but now we are choosing without replacement n of N elements contained in the set.

There are $\binom{M}{x}$ ways of choosing x of the M successes and $\binom{N-M}{n-x}$ ways of choosing $n-x$ of the $N-M$ failures and hence $\binom{M}{x}\binom{N-M}{n-x}$ ways of choosing x successes and $n-x$ failures. Since, there are $\binom{N}{n}$ ways of choosing n of the N elements in the set. (We assume that they are all equally likely)

Then the probability of “ x successes in trials”

$$\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

Definition 5.6

A random variable X has a hypergeometric distribution and it is referred to as a hypergeometric random variable if and only if its probability distribution is given by,

$$h(x;n,N,M) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

for $x = 0, 1, 2, \dots, n$ $x \leq M$ and $n-x \leq N-M$

Note:

Thus for sampling without replacement, the number of successes in n trials is a random variable having a hypergeometric distribution with the parameters n , N and M .

As part of an air-pollution survey, an inspector decides to examine exhaust of six of a company's 24 trucks. If four of the company's trucks emit excessive amounts of pollutants, what is the probability that none of them will be included in the inspector's sample?

$x=0$ $n=6$ $N=24$ $M=4$

$$h(0;6,24,4) = \frac{\binom{4}{0} \binom{24-4}{6-0}}{\binom{24}{6}} = 0.2880$$

Theorem 5.7: The mean and the variance of the hypergeometric distribution are,

$$\mu = \frac{nM}{N} \quad \text{and} \quad \sigma^2 = \frac{nM(N-M)(N-n)}{N^2(N-1)}$$

Note: When N is large and n is relatively small compared to N (n should not exceed 5 percent of N), there is not much difference between sampling without replacement, and the formula for the binomial distribution with the parameters n and $\theta = \frac{M}{N}$ may be used to approximate hypergeometric probabilities.

Among the 120 applicants for a job, only 80 are actually qualified. If 5 of the applicants are randomly selected for an in-depth interview, find the probability that only two of the five will be qualified for the job by using,

- (a) The formula for the hypergeometric distribution
- (b) The formula for the binomial distribution with $\theta = \frac{80}{120}$ as an approximation

Solution:

- (a) Substituting $x=2$, $n=5$, $N=120$, $M=80$

$$h(2;5,120,80) = \frac{\binom{80}{2} \binom{40}{3}}{\binom{120}{5}} = 0.164$$

- (b) $x=2$, $n=5$, $\theta = \frac{80}{120} = \frac{2}{3}$

$$b(2;5,\frac{2}{3}) = \binom{5}{2} \left(\frac{2}{3}\right)^2 \left(1 - \frac{2}{3}\right)^3 = 0.165$$

6. Multinomial distribution

An immediate generalization of the binomial distribution arises when,

1. Each trial has more than two possible outcomes.
2. The probabilities of the respective outcomes are the same for each trial.
3. The trials are all independent.

Example:

1. When performed interviewed by an opinion poll are asked whether they are for a particular candidate against him or undecided.
2. When sample of manufactured products are rated excellent, above average, average or poor.

In general, let us consider a case whether there are n independent trials permitting k mutually exclusive outcomes whose respective probabilities are $\theta_1, \theta_2, \dots, \theta_k$. ($\sum_{i=1}^k \theta_i = 1$).

Referring to the outcomes as being of the first kind, the second kind, and ... the k^{th} kind; we shall be interested in the probability of getting x_1 outcomes of the first kind, x_2 outcomes of the second kind,and x_k outcomes of the k^{th} kind. ($\sum_{i=1}^k x_i = n$)

Proceeding as in the derivation of the formula for the binomial distribution, we first find that the probability of the getting x_1 outcomes of the first kind, x_2 outcomes of the second kind,and x_k outcomes of the k^{th} kind in a specific order is,

$\theta_1^{x_1} \cdot \theta_2^{x_2} \dots \theta_k^{x_k}$. To get corresponding probabilities for that many outcomes of each kind in “any order” we have to multiply the probability for any specific order by,

$$\binom{n}{x_1 \ x_2 \ \dots \ x_k} = \frac{n!}{x_1! x_2! \dots x_k!} \quad (\text{According to the theorem 1.8})$$

Definition 5.8: The random variables $x_1, x_2, \dots, x_k$ have a multinomial distribution and they are referred to as multinomial random variables if and only if their joint probability distribution is given by,

$$f(x_1, x_2, \dots, x_k; n, \theta_1, \theta_2, \dots, \theta_k) = \binom{n}{x_1 \ x_2 \ \dots \ x_k} \theta_1^{x_1} \cdot \theta_2^{x_2} \dots \theta_k^{x_k}$$

for $x_i = 0, 1, 2, \dots, n$ for each i ,

$$\text{where } \left(\sum_{i=1}^k x_i = n \right) \text{ and } \left(\sum_{i=1}^k \theta_i = 1 \right)$$

Note: Thus, the number of outcomes of the different kinds are random variables having the multinomial distribution with parameters $n, \theta_1, \theta_2, \dots, \theta_k$.

Example: A certain city has three television stations. During on Saturday nights channel A has 50% of the viewing audience, channel B has 30% of the viewing audience and channel C has 20% of the viewing audience. Find the probability that among 8 television viewers in that city, randomly chosen on a Saturday night, 5 will be watching channel A, 2 will be watching channel B, and 1 will be watching channel C.

$$x_1 = 5, x_2 = 2, x_3 = 1 \quad \theta_1 = 0.5, \theta_2 = 0.3, \theta_3 = 0.2$$

$$n=8$$

$$f(5,2,1;8,0.5,0.3,0.2) = \frac{8!}{5! \cdot 2! \cdot 1!} (0.5)^5 (0.3)^2 (0.2)^1$$

$$= 0.0945$$

7. Poisson Distribution

Consider an application of binomial distribution when $x=18$, $n=3000$, $\theta=0.005$

$$b(18;3000,0.005) = \binom{3000}{18} (0.005)^{18} (0.995)^{2982}$$

In this section, we shall present a probability distribution that can be used to approximate binomial probabilities when $n \rightarrow \infty, \theta \rightarrow 0$ while $n\theta$ remains constant.

Letting this constant be λ , i.e $n\theta = \lambda$

$$\therefore \theta = \frac{\lambda}{n}$$

Hence, the limiting distribution becomes,

$$p(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Definition 5.7: A random variable X has a Poisson distribution and it is referred to as a Poisson random variable if and only if its probability distribution is given by,

$$p(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Then in the limit when $n \rightarrow \infty, \theta \rightarrow 0$ and $n\theta = \lambda$ remains constant, the number of successes is a random variable having a Poisson distribution with the parameter λ .

Poisson distribution will provide a good approximation to the binomial probabilities when $n \geq 20$ and $\theta \leq 0.05$. When $n \geq 100$ and $n\theta < 10$, the approximation will generally be excellent.

Example: If 2 percent of the books bound at a certain bindery have defective bindings, use the Poisson approximation to the binomial distribution to determine the probability that 5 of 400 books bound by this bindery will have defective bindings.

$$X = 5, n = 400, \theta = 0.02, \lambda = 400 \times 0.02 = 8$$

$$p(5;8) = \frac{8^5 e^{-8}}{5!} = \frac{(32768)(0.00034)}{120} = 0.093$$

In practice Poisson probabilities are seldom obtained by direct substitution into the formula.

Example: Records show that the probability is 0.00005 that a car will have a flat tire while crossing a certain bridge. Use the Poisson distribution to approximate the binomial probabilities that among 10000 cars crossing the bridge.

- (a) Exactly 2 will have a flat tire
- (b) At most 2 will have a flat tire

Solution

$$(a) \quad x=2, \lambda = 10000 \times 0.00005 = 0.5$$

$$p(x;0.5) = 0.0758$$

$$\begin{aligned}(b) \quad x = 0,1,2 \\&= p(0;0.5)p(1;0.5) + p(2;0.5) \\&= 0.6065 + 0.3033 + 0.0758 \\&= 0.9856\end{aligned}$$

Theorem 5.8: The mean and the variance of the Poisson distribution are given by,

$$\mu = \lambda \quad \text{and} \quad \sigma^2 = \lambda$$

Theorem 5.9: The moment generating function of the Poisson distribution is given by,

$$M_X(t) = e^{\lambda(e^t-1)}$$

Proof:

By definition,

$$M_X(t) = E[e^{xt}] = \sum_{x=0}^{\infty} e^{xt} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$$

When $\sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$ can be recognized as the Maclaurin's series of e^z with $z = \lambda e^t$ then,

$$M_X(t) = e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t-1)}$$

If we get differentiate $M_X(t)$ twice with respect to t ,

$$M'_X(t) = \lambda e^t e^{\lambda(e^t-1)}$$

$$M''_X(t) = \lambda e^t e^{\lambda(e^t-1)} + \lambda^2 e^{2t} e^{\lambda(e^t-1)}$$

So that $\mu'_1 = M'_X(0) = \lambda$

$$\mu'_2 = M''_X(0) = \lambda + \lambda^2$$

$$\begin{aligned} \sigma^2 &= \mu'_2 - \mu^2 \\ &= \lambda + \lambda^2 - \lambda^2 = \lambda \end{aligned}$$

Although the Poisson distribution has been derived as a limiting form of the binomial distribution, it has many applications that have no direct connection with binomial distributions.

Poisson can serve as a model for the number of successes that occur during a given time interval or a special region when,

1. The number of successes occurring in non-overlapping time intervals or regions is independent.

2. The probability of single success occurring in a very short time interval or in a very small region is proportional to the length of the time interval or size of the region.
3. The probability of more than one success occurring in such a short time interval falling such a small region is negligible.

Ex:

- No of telephone calls per hour received by an office
- No of typing errors per page
- No of bacteria in a given culture

When the average no of successes λ for the given time interval or specified region is known.

Example: The average number of trucks arriving on any day at a truck depot in a certain city is known to be 12. What is the probability that on a given day fewer than 9 trucks will arrive at this depot?

Let X be number of trucks arriving on a given day.

$X=12$

Note: If in a situation where the preceding conditions apply, successes occur at a mean rate of α per unit time or per unit region, then the number of successes in an interval of t units of time or t units of specified region is a Poisson random variable with the mean $X = \alpha t$. Therefore, the number of successes X, in a time interval of length t units or a region of size t units has the Poisson distribution.

$$p(x;\alpha t) = \frac{e^{-\alpha t} \cdot (\alpha t)^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Example: A certain kind of sheet metal has, on the average five defects for 10 square feet. If we assume a Poisson distribution, what is the probability that a 15 square feet sheet of the metal will have at least six defects?

$$\alpha = 5$$

Let X denotes the number of defects in a 15-square-feet sheet of the metal. Since unit of area is 10 square feet, we have,

$$\lambda = \alpha t = (5)(1.5) = 7.5 \text{ and}$$

$$\begin{aligned}\Pr(X \geq 6) &= 1 - \Pr(X < 6) \\ &= 1 - [0.0006 + 0.0041 + 0.0156 + 0.0389 \\ &\quad + 0.0729 + 0.1094] \\ &= 1 - [0.2415] \\ &= 0.7585\end{aligned}$$

